Politecnico di Torino

Automatic Domain Randomization for Robust Sim-to-Real Transfer

Implementation and Parameter Relevance Analysis

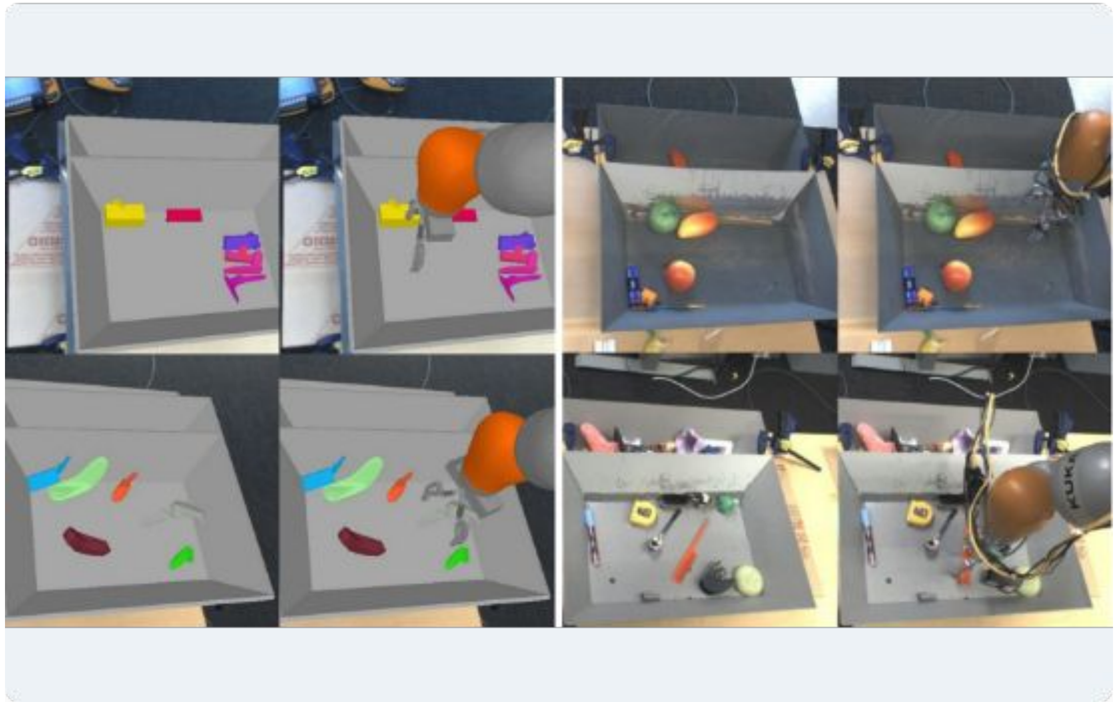
Luigi Marguglio

Marc'Antonio Lopez

Robot Learning

Introduction: The Reality Gap

- **Simulation:** Perfect physics, deterministic. The robot learns ideal gaits.
- **Reality:** Variable friction, sensor noise, unmodeled dynamics.
- **The Problem:** "Reality Gap" - The mismatch between simulated and physical dynamics causes failure in the real world.



The Solution: Domain Randomization (DR)

Concept

Instead of seeking a single perfect simulation, we vary physical parameters during training (Mass, Friction, etc.).

Goal

- The real world is treated as just "another sample" of the distribution.
- The agent learns invariance to parameter changes.
- **Robustness > Optimality**

Uniform Domain Randomization (UDR)

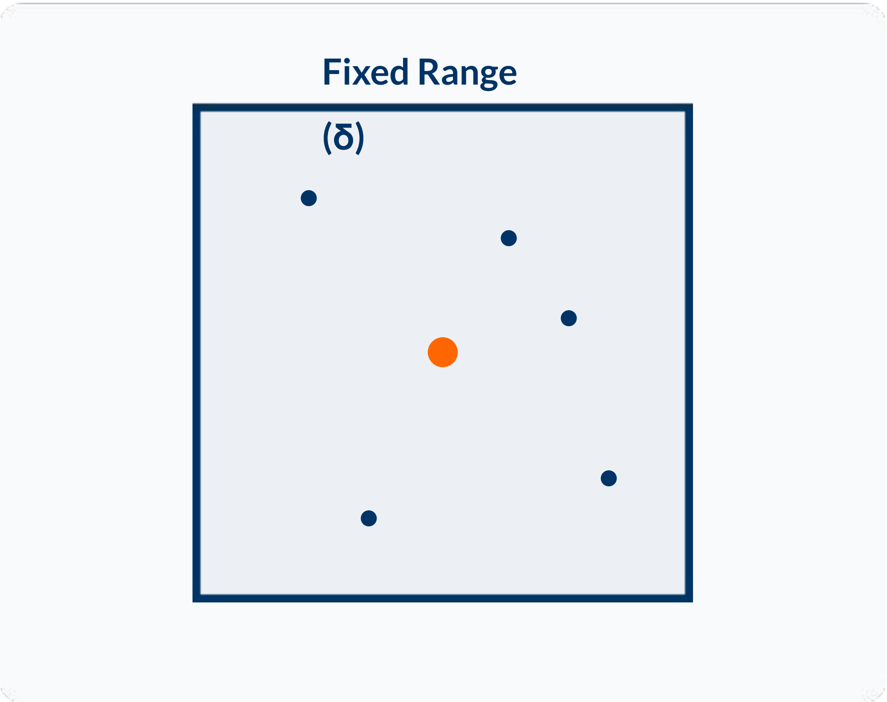
Mechanism

Parameters sampled from fixed uniform ranges defined *a priori*.

$$P \sim U(P_{nom} - \delta,\ P_{nom} + \delta)$$

Limitations

- **Fixed Boundary:** Agent never sees beyond the box.
- **Risk:** Ranges too narrow (no transfer) or too wide (impossible task).



Automatic Domain Randomization (ADR)

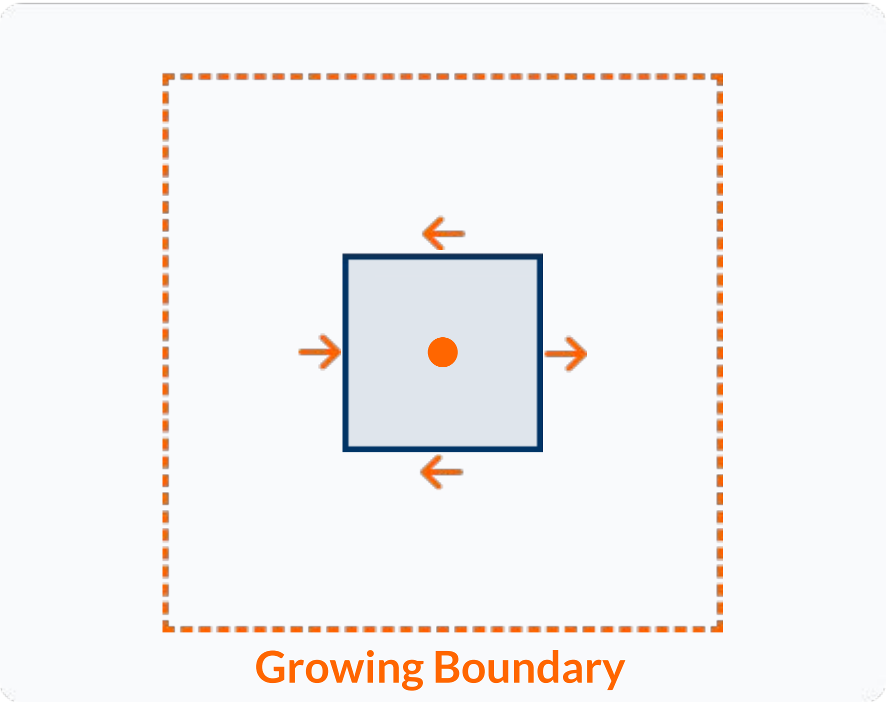
Adaptive Strategy

The parameter bounds are not fixed.

They **grow** as the agent gets better.

- **Perf ↑ :** Range Expands (Harder)
- **Perf ↓ :** Range Contracts (Easier)

Result: Automatic Curriculum



Paper Contributions

Part 1: ADR Evaluation

Systematic comparison of ADR against Baseline and UDR across multiple training durations (2.5M, 5M, 10M).

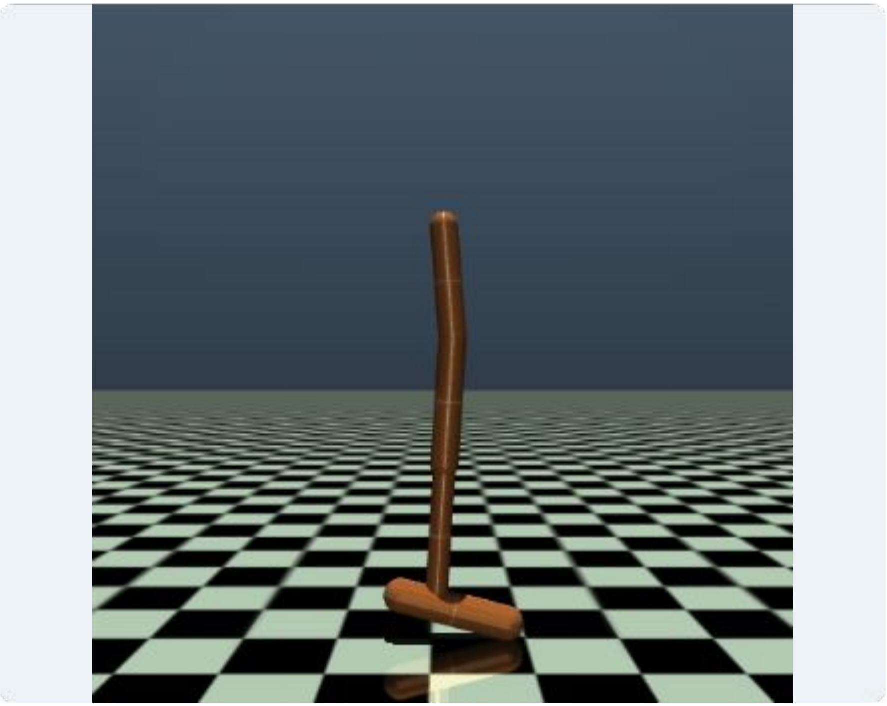
Part 2: Parameter Relevance

Ablation study analyzing the specific contribution of Mass, Damping, and Friction, including interaction effects.

Methodology: Environment

Setup

- **Agent:** Hopper (MuJoCo).
- **Source Env (Train):** Modified dynamics with -1kg torso mass offset (incorrect model).
- **Target Env (Test):** Nominal dynamics (Ground Truth).



ADR Implementation Details

State Space

Randomization ranges φ for Mass, Damping, Friction:

$$S = \{ \delta_m, \delta_d, \delta_f \}$$

Update Logic

Evaluated every N episodes:

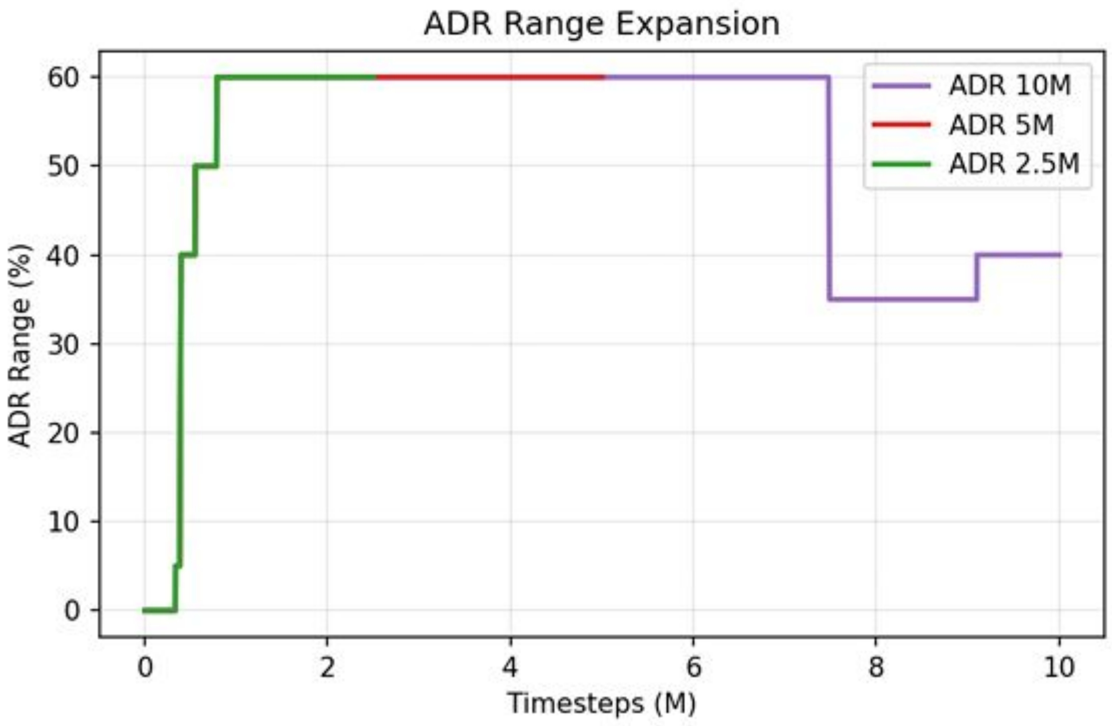
- **Reward ≥ 1200:** Expand Ranges (Difficult ↑)
- **Reward < 600:** Contract Ranges (Difficult ↓)

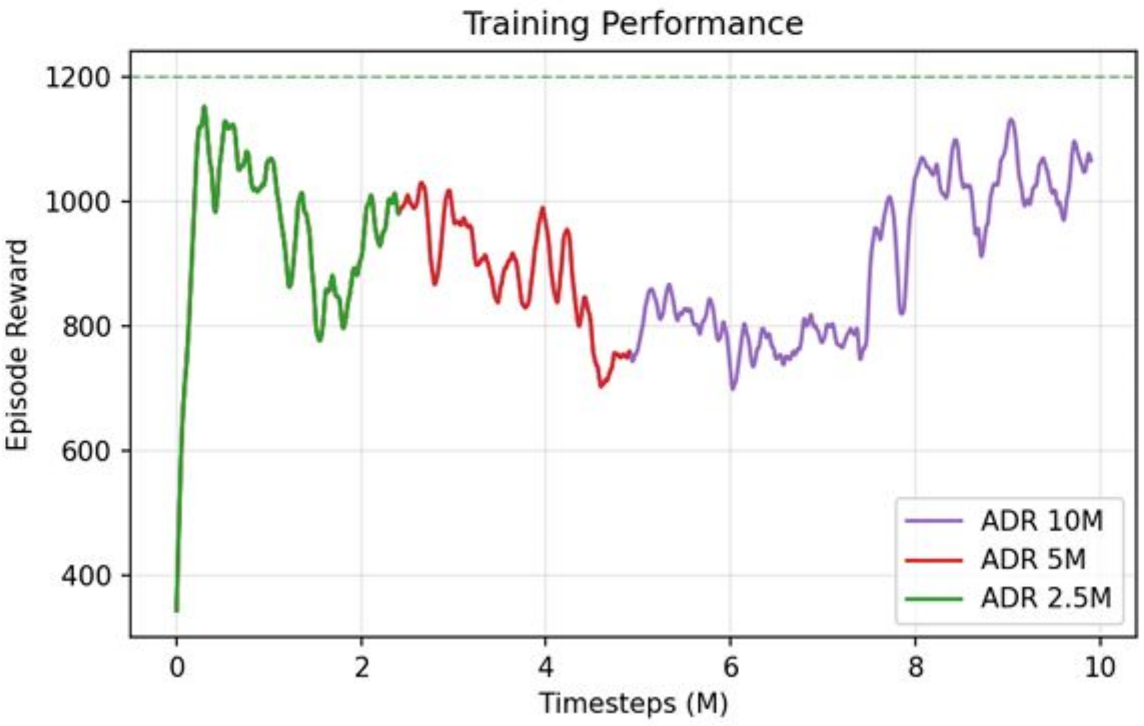
Ablation Study Design

Config Name	Mass	Damping	Friction
Baseline	-	-	-
UDR	✓ (Fixed)	✓ (Fixed)	✓ (Fixed)
adr_mass	✓	-	-
adr_damp	-	✓	-
adr_fric	**-**	**-**	**✓**
adr_mass_damp	✓	✓	-
adr_mass_fric	✓	-	✓
adr_damp_fric	-	✓	✓
adr_all	✓	✓	✓

Part 1 Results: Training Dynamics

- **Range Expansion:** 2.5M & 5M runs reach ~60% range, while 10M plateaus at ~40%.
- **Performance:** Longer training stabilizes the agent but does not necessarily force wider randomization ranges.





Part 1: Comparative Results

Evaluating the Sim-to-Real Gap (Source vs Target Performance).

Method	Gap	Result
Baseline	-34.2%	Severe failure in transfer.
UDR	+3.9%	Positive transfer, stable.
ADR 10M	**-0.4%**	Near-perfect stability.

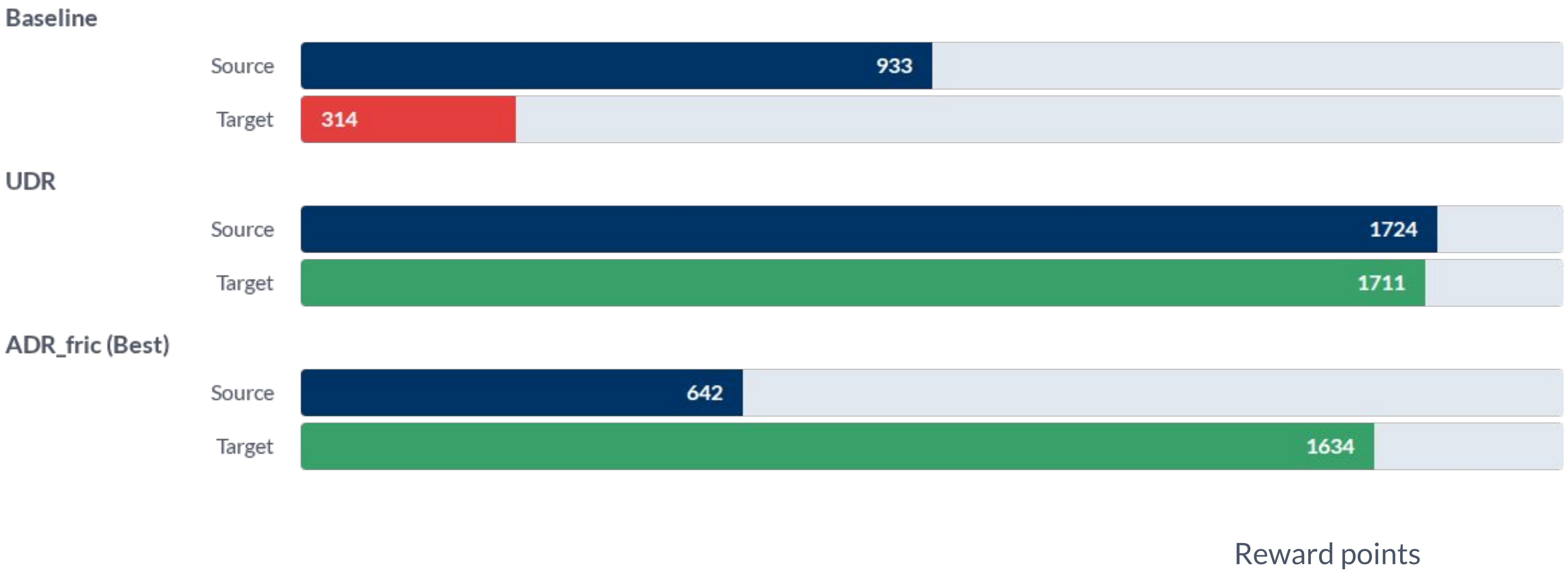
Part 2: Ablation Study Results

Ranking configurations by Transfer Gap.

Rank	Config	Gap Performance
1	**adr_fric**	**+154.6%**
2	adr_all	+14.1%
3	adr_damp	+8.0%
4	adr_damp_fric	+7.1%
5	udr	-0.8%
6	adr_mass_fric	-4.5%
7	adr_mass	-17.5%
8	adr_mass_damp	-31.7%
9	adr_none	-65.4%
10	baseline	-66.4%

Key Finding: Friction-only randomization vastly outperforms everything else.

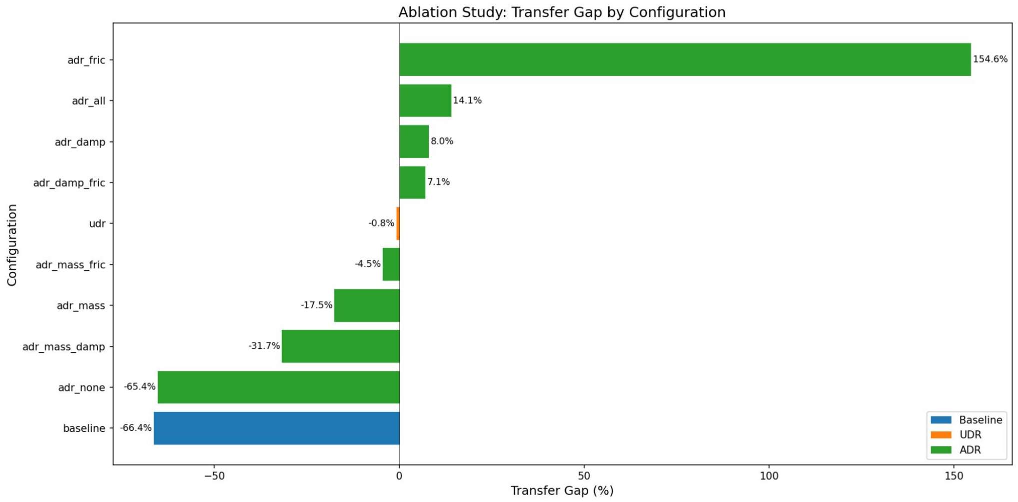
Visual Comparison: Source vs Target

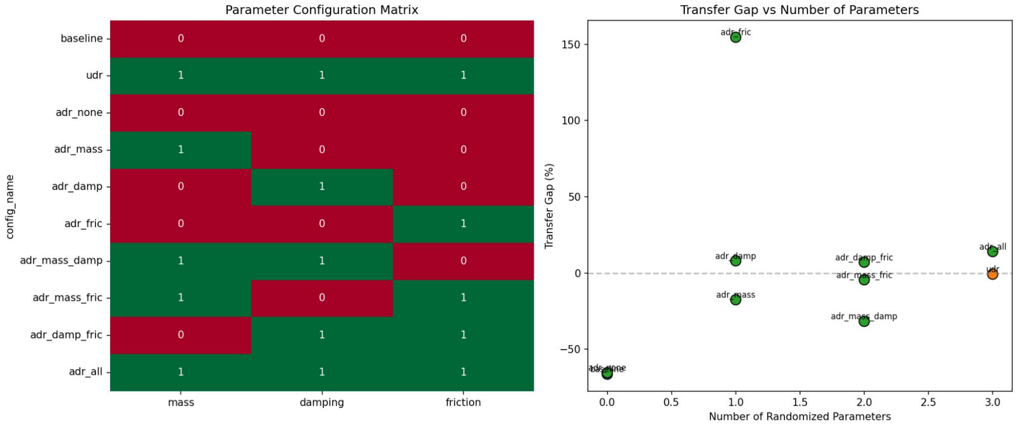


Transfer Gap by Configuration

Figure 3 & 5 Analysis

- **Non-Linearity:** More parameters != Better performance.
- **Complexity:** Increasing the number of randomized parameters does not monotonically increase the Transfer Gap.
- **Selective Wins:** Specific single parameters (Friction) outperform complex combinations (All).





Statistical Analysis: Marginal Contributions

Method

$$MC(p) = Avg(Gap_{with\_p}) - Avg(Gap_{without\_p})$$

Results

- **FRICTION: +68.7%** (p=0.07)
  The main driver.
- **MASS: -15.7%**
  Negative contribution.
- **DAMPING: -0.86%**
  Mostly neutral.

Interaction Analysis

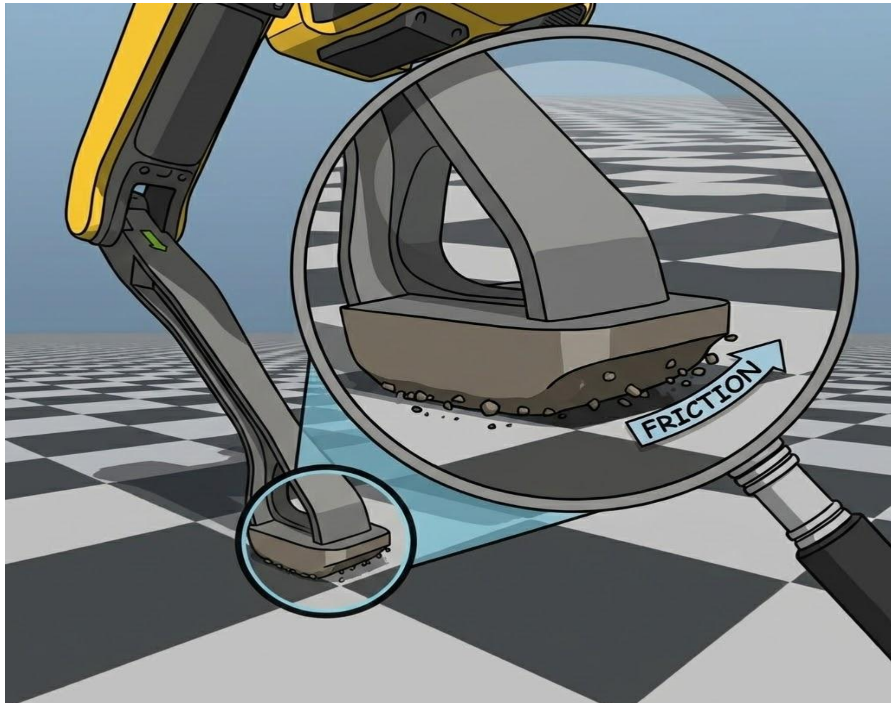
Do parameters work together or fight each other?

Interaction	Effect	Type
Mass × Friction	-94.5%	Strongly Antagonistic
Damping × Friction	-106.2%	Strongly Antagonistic
Mass × Damping	+5.0%	Synergistic

Adding Mass/Damping actively destroys the benefits of Friction randomization.

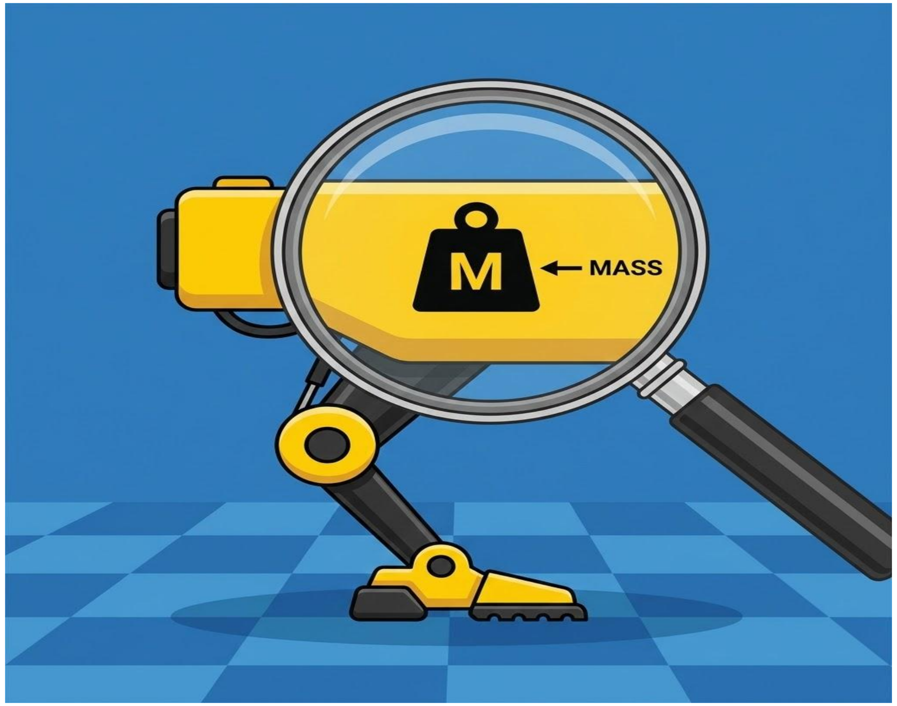
Discussion: Friction is King

- **Contact Dynamics:** Locomotion relies on foot-ground interaction. Errors here propagate instantly.
- **Misspecified:** Simulators often use simplified friction models.
- **Learning:** Randomizing friction forces the agent to learn stable stepping strategies.



Discussion: Mass Randomization can Hurt

- **Interference:** The Source Env already has a -1kg mass offset. Randomizing mass might mask this specific bias rather than fixing it.
- **Over-robustification:** The agent becomes too conservative to handle extreme masses that don't exist in the Target.



Practical Recommendations

Goal	Recommendation
Max Transfer	**adr_fric**
Balanced	adr_damp_fric
Conservative	UDR
Avoid	adr_mass, adr_mass_damp

Thank you

Marc’Antonio Lopez s336362@studenti.polito.it

Luigi Marguglio s337618@studenti.polito.it

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